

A Hilbert-Module Block Criterion for Coronas Failing BCP

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Source Problem

The source paper is Jinghao Huang, Karimbergenov Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Definitions

Let E be a Hilbert C^* -module. Write $\mathcal{L}(E)$ for its adjointable operators, $\mathcal{K}(E)$ for its compact operators, and

$$Q : \mathcal{L}(E) \longrightarrow \mathcal{L}(E)/\mathcal{K}(E)$$

for the quotient map.

For a finite set $F \subseteq \mathbb{N}$, write

$$e_F = \sum_{n \in F} e_n.$$

Definition 1. *A sequence (e_n) of nonzero pairwise orthogonal projections in $\mathcal{K}(E)$ is called a BCP block system for the corona $\mathcal{L}(E)/\mathcal{K}(E)$ if the following hold.*

1. Compact-vanishing. *For every $C \in \mathcal{K}(E)$ and every $\varepsilon > 0$, there is a finite set $F_0 \subseteq \mathbb{N}$ such that*

$$\|Ce_F\| < \varepsilon$$

whenever $F \subseteq \mathbb{N} \setminus F_0$ is finite.

2. Finite block equivalence. Whenever finite sets $F, G \subseteq \mathbb{N}$ have the same cardinality, there is a partial isometry $v \in \mathcal{L}(E)$ with

$$v^*v = e_F, \quad vv^* = e_G.$$

3. Strict gluing. If (F_k) and (G_k) are pairwise disjoint finite subsets of $\mathbb{N}$, $|F_k| = |G_k|$, and

$$v_k^*v_k = e_{F_k}, \quad v_kv_k^* = e_{G_k},$$

then there is $V \in \mathcal{L}(E)$, $\|V\| \leq 1$, such that

$$Ve_{F_k} = v_k, \quad k \geq 1.$$

4. Tail norm recovery. For every $T \in \mathcal{L}(E)$, every finite set $F_0 \subseteq \mathbb{N}$, and every $\varepsilon > 0$, there is a nonempty finite set $F \subseteq \mathbb{N} \setminus F_0$ such that

$$\|Te_F\| > \|Q(T)\| - \varepsilon.$$

Main Result

Theorem 1. *If the corona $\mathcal{L}(E)/\mathcal{K}(E)$ admits a BCP block system, then $\mathcal{L}(E)/\mathcal{K}(E)$ fails BCP.*

Corollary 1. *Let H be an infinite-dimensional separable Hilbert space and let D be a nonzero unital C^* -algebra. Then*

$$(\mathcal{B}(H) \otimes D)/(\mathcal{K}(H) \otimes D)$$

fails BCP.

Idea of the Proof

The block system is designed to isolate exactly the mechanism used in the stabilized Calkin proof. Tail norm recovery gives, for each proposed center, a late compact block on which that center nearly realizes its quotient norm. Compact-vanishing and finite block equivalence let us move a copy of that block to a fresh range where the finite-column compact operator Te_F is almost invisible. Strict gluing assembles these local partial isometries into one adjointable operator. Its corona class has norm one and remains outside every ball $B(\xi, \|\xi\|)$ from the proposed countable cover.

Proof

We use the standard equivalent form of BCP: if a normed space Y has BCP, then S_Y is covered by countably many balls $B(y_j, \|y_j\|)$.

Lemma 1 (fresh range escape). *Let (e_n) be a BCP block system. Let $F \subseteq \mathbb{N}$ be finite and nonempty, let $X \in \mathcal{K}(E)e_F$, let $F_0 \subseteq \mathbb{N}$ be finite, and let $\varepsilon > 0$. Then there are a finite set*

$$G \subseteq \mathbb{N} \setminus F_0, \quad |G| = |F|,$$

and a partial isometry $v \in \mathcal{L}(E)$ such that

$$v^*v = e_F, \quad vv^* = e_G,$$

and

$$\|v - X\| \geq \|X\| - \varepsilon.$$

Proof. By compact-vanishing applied to X^* , enlarge F_0 if needed so that

$$\|e_G X\| = \|X^* e_G\| < \varepsilon$$

for every finite $G \subseteq \mathbb{N} \setminus F_0$. Choose such a G with $|G| = |F|$, and choose v by finite block equivalence. Since $(1 - e_G)v = 0$,

$$\|v - X\| \geq \|(1 - e_G)(v - X)\| = \|(1 - e_G)X\| \geq \|X\| - \|e_G X\| > \|X\| - \varepsilon.$$

□

Proof of Theorem 1. Let

$$\xi_i = Q(T_i), \quad i \geq 1,$$

be an arbitrary countable family of centers in $\mathcal{L}(E)/\mathcal{K}(E)$, with $T_i \in \mathcal{L}(E)$. Put $r_i = \|\xi_i\|$. Choose a map $\pi : \mathbb{N} \rightarrow \mathbb{N}$ such that every value occurs infinitely often.

We recursively choose finite domain blocks F_k , finite range blocks G_k , and partial isometries v_k . Suppose the previous blocks have been chosen. Let H_{k-1} be the finite union of all previous F_j 's and G_j 's. Put $i = \pi(k)$. By tail norm recovery, choose a nonempty finite set

$$F_k \subseteq \mathbb{N} \setminus H_{k-1}$$

such that

$$\|T_i e_{F_k}\| > r_i - \frac{1}{k}.$$

Apply Lemma 1 to

$$F = F_k, \quad X = T_i e_{F_k},$$

and to a finite forbidden set containing $H_{k-1} \cup F_k$. We obtain

$$G_k \subseteq \mathbb{N} \setminus (H_{k-1} \cup F_k)$$

with $|G_k| = |F_k|$, and a partial isometry v_k satisfying

$$v_k^* v_k = e_{F_k}, \quad v_k v_k^* = e_{G_k},$$

and

$$\|v_k - T_i e_{F_k}\| > r_i - \frac{2}{k}.$$

By strict gluing, there is $V \in \mathcal{L}(E)$, $\|V\| \leq 1$, such that

$$V e_{F_k} = v_k, \quad k \geq 1.$$

We claim that $\|Q(V)\| = 1$. The upper bound is clear. For every $C \in \mathcal{K}(E)$, compact-vanishing gives

$$\|C e_{F_k}\| \rightarrow 0,$$

and hence

$$\|V - C\| \geq \|(V - C)e_{F_k}\| \geq \|v_k\| - \|C e_{F_k}\| = 1 - \|C e_{F_k}\|.$$

Taking $k \rightarrow \infty$ and then the infimum over $C \in \mathcal{K}(E)$ gives $\|Q(V)\| \geq 1$.

Fix i . If $r_i = 0$, then

$$\|Q(V) - \xi_i\| \geq 0 = r_i.$$

If $r_i > 0$, let $k \rightarrow \infty$ along the infinite set $\{k : \pi(k) = i\}$. For any $C \in \mathcal{K}(E)$,

$$\|V - T_i - C\| \geq \|(V - T_i - C)e_{F_k}\| \geq \|v_k - T_i e_{F_k}\| - \|C e_{F_k}\| > r_i - \frac{2}{k} - \|C e_{F_k}\|.$$

Letting $k \rightarrow \infty$ along this subsequence gives

$$\|V - T_i - C\| \geq r_i.$$

Taking the infimum over $C \in \mathcal{K}(E)$, we get

$$\|Q(V) - \xi_i\| \geq r_i = \|\xi_i\|.$$

Thus $Q(V) \in S_{\mathcal{L}(E)/\mathcal{K}(E)}$ is outside every ball

$$B(\xi_i, \|\xi_i\|).$$

The countable family of centers was arbitrary, so the corona fails BCP. $\square$

Proof of Corollary 1. Let $E = H \otimes D$ be the standard Hilbert D -module. Since D is unital,

$$\mathcal{L}(E) \cong \mathcal{B}(H) \otimes D, \quad \mathcal{K}(E) \cong \mathcal{K}(H) \otimes D.$$

Fix an orthonormal basis (h_n) of H , and let e_n be the projection onto the D -summand $h_n \otimes D$. Then $e_n \in \mathcal{K}(E)$, the compact-vanishing property follows from norm approximation by finite matrices in $\mathcal{K}(H) \otimes D$, and finite block equivalence is implemented by the usual matrix-unit partial isometries tensored with 1_D . Strict gluing is the strong operator sum of those partial isometries, tensored with 1_D .

It remains only to recall tail norm recovery. If f is a finite sum of coordinate projections, $T \in \mathcal{B}(H) \otimes D$, and $r = \|Q(T)\|$, then

$$r \leq \|T(1 - f)\|.$$

Represent D faithfully on a Hilbert space K_D . Choose a unit vector $\xi \in (1 - f)H \otimes K_D$ with $\|T(1 - f)\xi\| > r - \varepsilon/2$, and approximate ξ by a vector with finite H -support, close enough to preserve the weaker lower bound $r - \varepsilon$. The corresponding finite set F of coordinates is disjoint from the support of f , and

$$\|T e_F\| > r - \varepsilon.$$

Thus (e_n) is a BCP block system, so Theorem 1 applies. $\square$

Scope and Nonunital Obstruction

This packet is not merely a restatement of the unital stabilized-corona theorem. It isolates the exact projectional block mechanism behind that proof. The same argument would apply to any Hilbert-module corona once a BCP block system is verified.

For a nonunital coefficient algebra D , the standard coordinate projections on the Hilbert D -module $\ell_2(D)$ are generally multiplier projections, not compact module projections. Thus compact block projections and tail norm recovery are not automatic. This is the current obstruction to promoting the unital stabilized-corona theorem to arbitrary nonunital stable multiplier coronas.

Verification and Search Bounds

The run indexes were searched for arXiv:2311.14257 together with Hilbert module, block frame, tail recovery, orthogonal compact projections, multiplier corona, and BCP. No existing packet for this Hilbert-module block criterion was found. The proof is self-contained given the block-system axioms and uses only standard Hilbert-module notation.

Human Review Focus

Review Definition 1, especially whether strict gluing and tail norm recovery are stated at the right level of generality. The proof of Theorem 1 is then a direct diagonal escape argument.

References

- [1] Jinghao Huang, Karimbergenov Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.